

SMAS-1 | Assignment 3: Image Transformations

$A1 = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$ (Scaling)

(a) $T(e1) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}^T$, $T(e2) = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The image stretches to twice its width (first column scales x by 2) while compressing to half its height (second column scales y by 0.5).

$A2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90 Degree Rotation)

(a) $T(e1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T$, $T(e2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The x-axis maps to the positive y-axis and the y-axis maps to the negative x-axis, rotating the image 90 degrees CCW.

$A3 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (Horizontal Shear)

(a) $T(e1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$, $T(e2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The x-axis remains the same (first column), but the y-axis is sheared to the right by some factor proportional to its height (second column). This slants the image.

$A4 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (Reflection in y-axis)

(a) $T(e1) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}^T$, $T(e2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T$

(c) Rank = 2

(d) No dimension/information is lost.

(e) The x-coordinates are flipped to the opposite sign (first column is negated) while the y-coordinates remain the same, creating a mirror image across the y-axis.

$A_5 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (Projection onto x-axis)

(a) $T(e_1) = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$, $T(e_2) = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$

(c) Rank = 1

SMAS-1 | Assignment 3: Image Transformations

(d) Dimension/information IS lost. The 2D image collapses into a 1D space.

(e) Because the second column is all zeros, all pixels' y-values collapse to zero, making it a single horizontal line.